



# Mauro D'Achille

## Curriculum Vitae

### Education

- 2023–Present **PhD candidate**, *Friedrich-Schiller-Universität Jena, Germany*
- 2019–2023 **Master Degree in Theoretical Physics**, *Università degli studi di Trieste, Trieste*, Grade: 110/110 with honors
- 2015–2018 **Bachelor Degree in Physics**, *Sapienza Università di Roma, Rome*, Grade: 110/110
- 2010–2015 **Scientific High School**, *Giuseppe Peano – Monterotondo (RM)*, Grade: 100/100

### Master Thesis

- Title *Shear operator by means of Loop Quantum Gravity techniques*
- Supervisors Professor Daniele Pranzetti & Professor Stefano Ansoldi

### Bachelor Thesis

- Title *Il lensing gravitazionale – Gravitational lensing effect*
- Supervisor Professor Luca Lamagna

### Internships & Workshops

- 2022 **Fundamental Problems in Quantum Physics 2022**, *Università degli studi di Trieste, Trieste, IT*  
In this second school on quantum foundations dedicated to professor GianCarlo Ghirardi, it was discussed some of the most relevant topics addressing foundational questions about Quantum Mechanics. Some of the main topics were Collapse Models, Bohmian Mechanics, macroscopic superposition and Quantum Information.
- 2022 **The International Conference Loops 2022**, Lyon, FR  
The scientific program of these conferences was to highlight the recent developments in quantum gravity and cosmology.
- 2022 **Loops 2022 Summer School**, THE QUANTUM GRAVITY TEAMS OF THE CPT MARSEILLE AND THE ENS LYON, Marseille, FR  
I participated to this one week summer school on Loop Quantum Gravity in order to go deeper in the understanding of my master thesis topics.

2020 **ESOF 2020 – Quantum Communications Project**, WITH CNR OF FLORENCE AND PROFESSOR ANGELO BASSI, Trieste, IT

I participated as a developer of the error correction software and the real-time visualisation software for the first public demonstration of communication via optical fibres encrypted with quantum technologies in Italy.

2019 **Quantum Software on Real Quantum Computers**, IBM – HANDS-ON WORKSHOP, Università degli studi di Trieste, Trieste, IT

Taught by Michele Grossi, this two-days workshop was an introduction of quantum gates with related tutorials. In particular, I learnt how to implement some quantum algorithms on the IBM Q platform using qiskit.

## Work Experience

2018 – **Undergraduate and High School Tutor**, IN MATHEMATICS AND PHYSICS, Italy  
Present Learnt to adjust my teaching style to each different student and to their needs.

## Computer Skills

Advanced Programming language: C, Fortran, Python, Mathematica  
Intermediate Programming language: R, Swift

## Languages

Italian Mothertongue  
English Advanced B2

*written and oral*

## Interests

- Sport: 2nd Dan Black Belt in Wado-Ryu Karate and Former Competitive Athlete
- Music: Trumpet
- Travelling
- Reading

## European Driving License: B